CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA)
Semester-II University Examination 2009-10
CA705 Advanced Programming and Data Structure

Date: 17.05.10, Monday **Time: 12.30 p.m to 03.30 p.m** **Maximum Marks: 70**

Instructions:

1. **Attempt the two sections in separate answer books.**
2. **Figure to the right indicate marks.**
3. **Make suitable assumptions wherever necessary and state them.**

SECTION-1

Q1 Answer the following questions. **[11]**

(a) **Differentiate following terms.** 4
(i) Call by Address **Vs.** Call by Value
(ii) Formal Parameter **Vs.** Actual Parameter

(b) Explain storage classes available in C. 4

(c) Define: modular programming, function prototype, recursion. 3

Q2 Answer the following question. **[12]**

(a) What is dynamic memory allocation? What are the advantages of it? 4

(b) Differentiate: 4
1. Array of pointer **Vs.** pointer to array.
2. Structure **Vs.** union.

(c) How to initialize array of structure, explain with example. 4

OR

(a) Explain function available in 'C' for dynamic memory allocation 4

(b) 1. Differentiate: Array and structure. 2
2. What is mean by bit field? Give syntax to declare it. 2

(c) Explain different ways to pass structure in function with example. 4

Q3 Answer the following questions. **[12]**

(a) What is mean by file? What are advantages of file compare to IO.List out different opening mode of file. 4

(b) Write short note on command line arguments. 4

(c) Explain different functions which are responsible for random access to files. 4

OR

(a) Explain different functions which are responsible for error handling. 4

(b) What is macro? Explain different types of preprocessor statements. 4

(c) Write a program to count no of characters, words, lines for a file passed as a command line argument. 4

SECTION-2

Q4 Answer the following questions. **[11]**

(a) Compare array **Vs.** linked list with specialized features. 3

(b) **Do as directed.**
1. Differentiate: Singly Circular List **Vs.** Doubly Circular List. 2
2. State weather following statements is true or false with justification. 2
(a) Both kind of access, sequential as well as random, is possible Using linked list.
(b) Insertion and deletion of an element is effective in array than linked list.

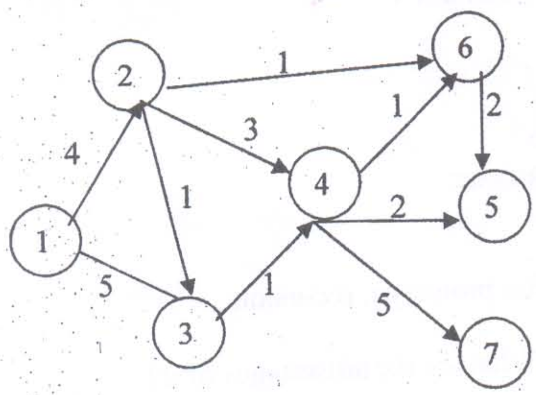
(c) Write an algorithm to delete and display element of doubly linked list. 4

[P.T.O]

Q5		**Answer the following questions.**	**[12]**
	(a)	Explain with example, different representation of graph.	**4**
	(b)	What is mean by Graph Traversal? Explain BFS Algorithm with proper example.	**6**
	(c)	Define following terms : Isomorphic Graph, Connected Graph	**2**

OR

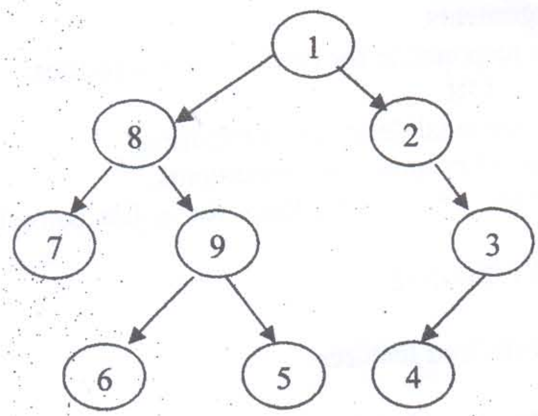
Q5		**Answer the following questions.**	**[12]**
	(a)	Trace Dijkstra's algorithm for following graph.	**5**



	(b)	Explain minimum cost spanning tree algorithm with example	**5**
	(c)	Define following terms: Degree of graph, Parallel edges.	**2**
Q6		**Answer the following questions.**	**[12]**
	(a)	What is mean by Hash Table? Give different ways to generate hash function?	**4**
	(b)	What is mean by traversal of tree? Explain in order traversal algorithm with example.	**6**
	(c)	Define following terms : Binary Tree, Height of Tree	**2**

OR

	(a)	Differentiate: Linear Open Addressing **Vs.** Chaining method. What is mean by collision? Discuss different ways to remove collision in linear open addressing.	**6**
	(b)	Traverse following tree using a) in order b) preorder c) post order traversal method.	**4**



	(c)	Define following terms: Complete binary tree, Directed tree.	**2**